Started on	Saturday, 10 October 2026, 11:10 PM
State	Finished
Completed on	Saturday, 10 October 2026, 11:17 PM
Time taken	7 mins 32 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
int main(){
    int n, val;
    if(scanf("%d", &n)!=1) return 0;
    int visited[n+1];
    for(int i=0;i<=n;i++) visited[i]=0;
    for(int i=0;i<n;i++){
        if(scanf("%d", &val)==1){
            if(visited[val]==1){
                printf("%d\n", val);
                return 0;
            }
            visited[val]=1;
        }
    }
    return 0;
}
```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Started on	Saturday, 10 October 2026, 11:37 PM
State	Finished
Completed on	Saturday, 10 October 2026, 11:40 PM
Time taken	3 mins 17 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```c
#include<stdio.h>
int main(){
    int n,val;
    if(scanf("%d", &n)!=1)return 0;
    int vis[n+1];
    for(int i=0;i<=n;i++) vis[i]=0;
    for(int i=0;i<n;i++){
        scanf("%d",&val);
        if(vis[val]==1){
            printf("%d\n",val);
            return 0;
        }
        vis[val]=1;
    }
    return 0;
}
```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Started on	Saturday, 10 October 2026, 11:19 PM
State	Finished
Completed on	Saturday, 10 October 2026, 11:30 PM
Time taken	10 mins 58 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1	10 57

1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
void solve(){
    int n1,n2,i=0,j=0,f=1;
    scanf("%d",&n1); int a[n1];
    for(int k=0;k<n1;k++) scanf("%d",&a[k]);
    scanf("%d",&n2); int b[n2];
    for(int k=0;k<n2;k++) scanf("%d",&b[k]);
    while(i<n1&&j<n2){
        if(a[i]<b[j])i++;
        else if(a[i]>b[j]) j++;
        else{
            printf(f?"%d":" %d",a[i]);
            f=0;i++;j++;
        }
    }
    printf("\n");
}
int main(){
    int t;
    if(scanf("%d", &t)==1)
        while(t--) solve();
    return 0;
}
```

	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Started on	Saturday, 10 October 2026, 11:41 PM
State	Finished
Completed on	Saturday, 10 October 2026, 11:49 PM
Time taken	7 mins 19 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1	10 57

1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
void solve(){
    int n1,n2,i=0,j=0,f=1;
    if(scanf("%d",&n1)!=1) return;
    int a[n1];
    for(int k=0;k<n1;k++) scanf("%d",&a[k]);
    if(scanf("%d",&n2)!=1) return;
    int b[n2];
    for(int k=0;k<n2;k++) scanf("%d",&b[k]);
    while(i<n1&&j<n2){
        if(a[i]<b[j])i++;
        else if(a[i]>b[j])j++;
        else{
            printf(f?"%d":" %d",a[i]);
            f=0;i++;j++;
        }
    }
    printf("\n");
}
int main(){
    int t;
    if(scanf("%d", &t)==1)
        while(t--)solve();
    return 0;
}
```

	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Started on	Saturday, 10 October 2026, 11:31 PM
State	Finished
Completed on	Saturday, 10 October 2026, 11:36 PM
Time taken	4 mins 19 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1

Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
int main(){
    int n,k;
    if(scanf("%d",&n)!=1)return 0;
    int a[n];
    for(int i=0;i<n;i++){
        scanf("%d",&a[i]);
    }
    if(scanf("%d", &k)!=1) return 0;
    for(int i=0;i<n;i++){
```

```
        for(int j=0;j<n;j++){
            if(i!=j&&a[j]-a[i]==k){
                printf("1\n");
                return 0;
            }
        }
    }
    printf("0\n");
    return 0;
}
```

	Input	Expected	Got	
✔	3 1 3 5 4	1	1	✔
✔	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✔
✔	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✔
✔	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Started on	Saturday, 10 October 2026, 11:49 PM
State	Finished
Completed on	Saturday, 10 October 2026, 11:54 PM
Time taken	4 mins 21 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1

Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
int main(){
    int n,k;
    if(scanf("%d", &n)!=1)return 0;
    int a[n];
    for(int idx=0;idx<n;idx++){
        scanf("%d",&a[idx]);
    }
    if(scanf("%d",&k)!=1)return 0;
    int i=0,j=1;
```

```
    while(i<n&&j<n){
        if(i==j){
            j++;
            continue;
        }
        int diff=a[j]-a[i];
        if(diff==k){
            printf("1\n");
            return 0;
        } else if(diff<k){
            j++;
        }else{
            i++;
        }
    }
    printf("0\n");
    return 0;

}
```

	Input	Expected	Got	
✔	3 1 3 5 4	1	1	✔
✔	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✔
✔	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✔
✔	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.